MATH2056 MT make up exam-SPR2021 Date:

The books and GSM phones are not allowed. A cheat sheet (single A4 page) and the calculators can be used.

All the answers must be clearly stated, otherwise no partial credit will be given.

Please write your answers well-ordered and clearly-visible: this is the responsibility of the student.

The duration is 110 minutes.

You can send the scanned answer sheets to my email: alper.sisman@marmara.edu.tr

1. **Ax=b & Vector spaces**

1.1. (10 pts)Write a matrix A that meets the following conditions:

- All entries are the integer numbers greater then 1.
- the matrix has 'm rows' and 'n columns', where n>3 and m>2
- Ax=b sometimes has infinite solutions
- Row space and Null space (for matrix A) have the same dimensions
- Column space spans a space that has the half of 'm' dimensions.

2. consider the matrix A=$\begin{bmatrix} 1 & \propto & -\gamma & \propto \\ 3\propto & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \propto^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$. Some of the elimination matrices are given below:

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -9 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$.

a. (10 pts) Find the unknown entries $\propto, \gamma, \beta, \mathrm{k}$ and $\mathrm{m} =?$

b. (10 pts) Find the eliminated upper triangle matrix U=? (If you couldn't find the variables in (a), all of them are one).

c. (15 pts) Determine the remaining elimination matrice(s).

3. a. Determine the inverse of the matrix A if it exists.

A=$\begin{bmatrix} 1 & 3 & 2 & -1 \\ 2 & 4 & 3 & 2 \end{bmatrix}$ (20 pts).

b. If it exists, how many different inverse can be found for this case? Does the space of possible inverses form a subspace? Comment on the dimension of the space (10 pts).

4. A geometry in 'k' dimensional space is given by x=5y expression, where 'x' and 'y' represents the axis. The intersection with another geometry gives a volume.

a. Give the minimum dimension of the space with your valuable comments (10 pts).

b. Propose an expression for the unknown geometry to have a volume in intersection (10 pts).

c. I proposed a geometry x+4y-z=6. Find all vectors in the intersection. Do these vectors form a vector space? The intersection has how many dimensions inside ow many dimensional space? (15 pts)